Kaivalya Swami

Software Engineer – AI & Search

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Education

2017–2021 **B.Tech, Computer Science & Engineering**, *Indian Institute of Technology Delhi, GPA – 8.5/10.*

Thesis Low-Resource Neural Machine Translation for 12 Indic Languages (Prof. Parag Singla). Fine-tuned mBART and created a standardised multilingual translation benchmark.

Ranks JEE Advanced 2017 – AIR **561** / 1.5M applicants | KVPY Scholar – AIR 571 (IISc Bangalore)

Olympiads National Astronomy Olympiad – Top 200, selected for INAO | Physics Olympiad – Top 1% in state

Experience

2021–Present **Software Engineer, AI Platform**, *DevRev*, Bengaluru.

Employee #40 – joined in stealth; now a unicorn with 800+ employees.
B2B SaaS platform with AI-native CRM, support agents, and enterprise search.
Core member of small team of AI/Search team owning retrieval, ranking, and agent infrastructure.

Reranking & Retrieval Pipeline

Two-stage ranking: RRF over BM25 + semantic retrieval, then cross-encoder reranker. Evaluated multiple Qwen-3-based reranking models; selected ZeRank-1-Small for best latency/quality trade-off – **+15% Recall@5** averaged across object types. Versioned ML configs for enabling staged rollouts. Powers all AI agents.

Search – Core Skill for AI Agents

Own search as the primary retrieval skill that all customer-facing AI agents invoke. Redesigned the skill contract – optimised query construction, snippet extraction, and result packaging – reducing average tool calls per query from **6.5** → **3.8** (**43% fewer skill calls**). Built *Compass*, a semantic clustering pipeline (inspired by Anthropic's *CLIO* and *How People Use ChatGPT*) to understand real user query patterns and feed insights back into skill contract tuning.

Agent Config Versioning & Knowledge Scoping

Configuration and knowledge-scoping framework for customer-facing AI agents – controlling which knowledge bases, tools, and retrieval strategies each agent accesses. Per-customer customisation with versioned, auditable configs.

A/B Testing Framework

Originally built to roll out reranking strategies on live traffic; generalised into a company-wide A/B testing framework adopted by multiple teams. Integrates with ML config system to test ranking strategies, reranker models, and retrieval parameters with measurable outcomes.

Analytics & Observability

End-to-end event instrumentation: CDC → Kafka → BigQuery data warehouse → LookerStudio dashboards for Search CTR, MRR, agent feedback across all customer orgs. Primary data source for PMs, CX, and Sales.

Also Owned ML inference service (Python/Go): async rewrite, 50%+ memory-leak reduction, all CVEs resolved. Shipped cross-lingual search to GA. Fixed distributed trace propagation across 3 microservices via gRPC header forwarding.

May–Jul 2020 **Data Engineering Intern**, *APT Portfolio.*

End-to-end alerting pipeline: Kafka message routing, C++/Python producer wrappers, MongoDB, Flask dashboard.

May–Jul 2019 **Research Intern**, ***Nanyang Technological University, Singapore***, Prof. Weichen Liu.

Neural network quantization – benchmarked Binarized NN (BMXNet) and XNOR-Net speedup on AlexNet & VggNet.

Dec 2019–2020 **Core Member**, *Tensor Dynamics (IIT Delhi Startup).*

Solar irradiance forecasting 3 hrs ahead using optical-flow cloud tracking on satellite imagery.

Publication

- K. Sharma, **K. Swami**, A. Shete, S. Husain, "Can Greenbergian universals be induced from language networks?", *SyntaxFest 2019*.

Teaching & Leadership

2020–21 **Teaching Assistant, DSA**, *IIT Delhi*, Prof. Rahul Garg.
Lab sessions for 20 students; assignment prep and exam grading.

2018–19 **Executive, Dev Club**, *IIT Delhi*.
Built intranet file-sharing service; lectured on Bash & Git to 100+ students.

Technical Skills

Languages	Python, Go, C++, SQL, Bash
ML & AI	PyTorch, LangChain, LlamaIndex, LangGraph, Hugging Face
Infrastructure	Kafka, BigQuery, MongoDB, gRPC, Docker, Kubernetes, Datadog, Langfuse, AWS, Lookerstudio
AI-Augmented	Cursor + Claude daily driver. Will learn your entire codebase before the onboarding call ends :)

Interests

Juggling	**Represented India at the European Juggling Convention (EJC) 2024** – the sole attendee from India.
House Dance	House dancer, have trained for 3 years. Love to learn all street style dance forms.
Racquet Sports	**Trained in table tennis and tennis and badminton**; avid player of all racquet sports.
Nature	Avid admirer of nature with a curious mind. Love to talk the cosmos and learn more and more.